

Kaylee Y. Vo

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EDUCATION

Harvard University

Master of Liberal Arts (ALM), Data Science | GPA: 4.0

Stanford University

Bachelor of Arts in Economics (BA) | *Minor: Statistics*

WORK EXPERIENCE

Nuna Inc., San Francisco, CA

Mar '23 – Jun '24

Software Engineer

- Productionized and maintained the Chronic Illness and Disability Payment System (CDPS) model. The model was deployed to Nuna's EMR Spark clusters, using Papermill for notebook execution, Docker for containerization, and Prefect for task orchestration. Optimized the model by reimplementing the entire algorithm, resulting in an 80% reduction in runtime.
- Built data transformations to map external client data to Nuna's data schema. Schemas are defined using Avro files and written to an AWS Glue data catalog. Performed Spark optimizations to reduce the runtime of the ETL pipeline. Wrote unit and integration tests to ensure data quality. The work culminated in a production Glue database used across the company.

Carbon Health, San Francisco, CA

Nov '22 – Jan '23

Senior Data Scientist

- Built and maintained DBT transformations to surface metrics to Looker dashboards, used by product, sales teams, and providers. Transformations were built on top of a Snowflake data warehouse.

Pngme, San Francisco, CA

Nov '21 – Nov '22

Data Scientist - Data Platform

- As the sole Data Scientist, spearheaded all Pngme's data initiatives. Created quarterly roadmaps and OKRs, planned bi-weekly sprints, and collaborated with internal and external stakeholders for GTM strategy and product feedback. Managed two reports (data science contractors) in addition to individual contributor work.
- Developed an MVP loan default prediction model, a logistic regression fitted on cross-sectional data. Features are financial metrics (i.e., average account balance, count of defaults) transformed into weight-of-evidence features using the OptBinning library. Lift, profit curves, and Shapley values were used as monitoring metrics, and AUC was the performance metric.
- Lead feature engineering and modeling R&D work to ensure ML model performance increases commensurately with data growth. In addition to the above numeric features, we generated custom word embeddings with BERT using user SMS data. The final model was a stacked ensemble of bi-directional LSTM, XGBoost, and an FCNN with logistic regression meta-model. The model was deployed to [production](#) and utilized by the client base to underwrite loans.

Toast Inc., Boston, MA

Nov '18 – April '20

Data Scientist - Toast Capital

- Built a new line of business (LOB) called Toast Capital. The product comprises three patented models: LSTM Revenue Prediction ([#11562425](#)), a probability of default model ([#11100575](#)), a dynamic pricing engine ([#10956974](#)), an ETL data pipeline, and an API endpoint that deploys JSON artifacts to internal teams. Total origination is >\$15 million, and profits are only second to the core LOB.
- Designed prototypes to improve the PD model. This includes model selection, model stability, model parsimony, TTC PD vs PIT PD, and testing regression assumptions. I also led an R&D initiative to develop a generalized PD model, which can generate PD estimates for any arbitrary time horizon.

SKILLS AND AWARDS

Computer Languages:	Python (Sklearn, PyTorch, Tensorflow), R, SQL (Postgres, Snowflake, BigQuery), bash, MATLAB, JAX
Mathematics and AI:	Vision (semantic segmentation, R-CNNs, ViT), LLM (LLaMA, BERT), Diffusion (U-Net, StableDiffusion), Multimodal (CLIP), Stochastic & Partial Differential Equations, Real & Functional Analysis, Matrix Theory, Causal Inference, Time Series.
Enterprise:	AWS (Sagemaker, S3, EC2, EMR, Glue), Prefect, Docker, Git/Github Actions, Kubernetes, Spark
Awards:	Stanford University Academic Scholarship (Full-Ride), Stanford Economics Student of the Month (10/15/15 – 12/15/15), Undergraduate Grant Recipient for Economics Research, Stanford Haas Center Research Grant Recipient, Center on Poverty & Inequality Certificate (1st Undergraduate Research Fellow), President of Stanford Economics Association

